Product recommendations & customer groups

GiftLane | Decision memo | 13 October 2026

To:	Sarah Okafor, Head of E-Commerce	**From:**	Data & ML Team
Subject:	Pilot results and next steps	**Status:**	Pilot, not deployment-ready

Recommendation

Keep best-selling products as the default recommendation box for now. A compressed recommendation model is fast and compact, but it did *not* improve accuracy on future purchases. Its customer-group map is promising for marketing exploration, subject to validation.

What we built and why

We analyzed **1,067,371** invoice lines from the 2009–2011 Online Retail II dataset. After removing cancellations, returns, invalid prices, service codes, and transactions without customer IDs, **801,559** lines remained. We used earlier purchases to construct a binary customer–product matrix (**5,473 customers** $\times$ **3,684 products**; **97.95% zeros**) and reserved the final two months for testing future, previously unseen items for **1,913** eligible customers.

We selected **Truncated SVD with 50 components**: unlike PCA, it processes the sparse purchase matrix without centering and gives compact customer and product representations for matching similar items.

Measured results

Method	Hit rate at 10	Difference vs. baseline
Top 10 best sellers	**26.87%**	—
Truncated SVD ($k = 50$)	24.05%	-2.82 points
PCA ($k = 50$)	24.10%	-2.77 points

A *hit* means at least one of a customer's genuinely new test-period products appears in the 10 suggestions. SVD performed slightly below PCA and both trailed the simple best-seller list.

Size and speed. SVD factors occupy **1.83 MB**, versus **161.30 MB** for an equivalent dense matrix (about **88$\times$ smaller**); the original sparse matrix already occupies just **3.33 MB**. On this computer, model fitting, item-neighbor updates, and scoring the test customers together took **0.129 seconds**. This *excludes* Excel loading and disk I/O, so it is not a complete nightly-job benchmark.

What compression loses. Keeping only 50 patterns blurs individual purchase combinations: the model has a **0.832 relative reconstruction error**. More components recover more detail but increase storage and computation.

Marketing view and next steps

The two-dimensional customer map suggests four descriptive groups: **occasional mixed-gift shoppers** (4,026), **lunch-bag and partyware shoppers** (510), **broad-catalog gift shoppers** (145), and **home-decor regulars** (792). Marketing could test seasonal assortments, party bundles, cross-category offers, and decor collections, respectively. These labels are hypotheses based on buying patterns, not confirmed customer preferences.

Risks: new customers and rare products have little history; missing customer IDs were excluded; and 2009–2011 data, especially a late-year test window, may not represent today's shopping. **Next:** retain best sellers in production, evaluate hybrid/personalized ranking on a separate validation period, and test any improvement before launch. Share only segment-level outputs, never customer IDs.